CS 506: Introduction to Quantum Computing

Quantum Fourier Transform Circuits and the No-Cloning Theorem

Scribe: Yaroslav Popryho

October 19, 2026

1 Overview

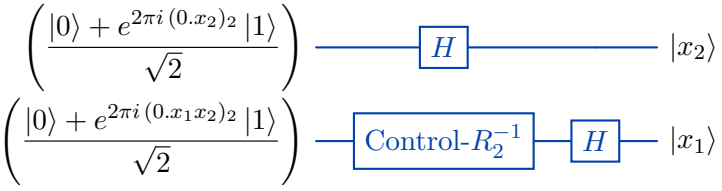
This lecture builds the quantum circuit for the Quantum Fourier Transform (QFT). Starting from the state $|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i\, wy} |y\rangle$ with $w = (0.x_1x_2\cdots x_n)_2$, we write $|\psi\rangle$ as a tensor product of single-qubit states and design a circuit C, built from Hadamard and controlled-R_k^{-1} gates, that maps it back to $|x_1x_2\cdots x_n\rangle$: first for $n = 2$ and $n = 3$, then for general n. We count the gates and the depth of this circuit, obtain QFT_{2^n} as the inverse of C, and contrast it with the QFT of order $m \neq 2^n$, which can only be computed approximately. We then recall the continuous and discrete Fourier transforms, and close with the no-cloning theorem of Wootters and Zurek.

2 Circuit for the Quantum Fourier Transform

2.1 The case $n = 2$

$n = 2, \quad w = (0.x_1x_2)_2$

$$\text{State} \quad |\psi\rangle = \frac{1}{2}\sum_{y=0}^{3} e^{2\pi i\, wy} |y\rangle \;=\; \left(\frac{|0\rangle + e^{2\pi i\,(0.x_2)_2}|1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + e^{2\pi i\,(0.x_1x_2)_2}|1\rangle}{\sqrt{2}}\right)$$



2.2 The case $n = 3$: product form and gate sequence

$n = 3$

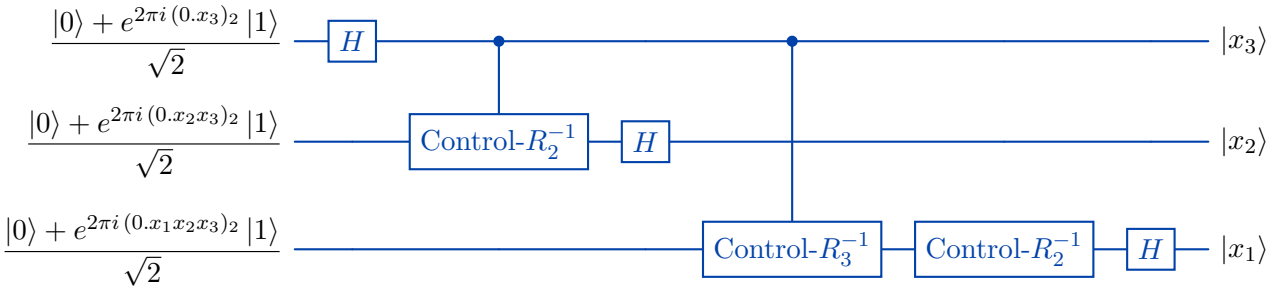
$$|\psi\rangle = \frac{1}{\sqrt{2^3}} \sum_{y=0}^{7} e^{2\pi i\, wy} |y\rangle$$

$$\frac{|0\rangle + e^{2\pi i\,(0.x_3)_2}\,|1\rangle}{\sqrt{2}} \quad \otimes \quad \frac{|0\rangle + e^{2\pi i\,(0.x_2x_3)_2}\,|1\rangle}{\sqrt{2}} \quad \otimes \quad \frac{|0\rangle + e^{2\pi i\,(0.x_1x_2x_3)_2}\,|1\rangle}{\sqrt{2}}$$
$$\Downarrow\ H \qquad\qquad \Downarrow\ \text{control-}R_2^{-1} \qquad\qquad \Downarrow\ \text{control-}R_3^{-1}$$
$$|x_3\rangle \qquad\qquad \Downarrow\ H \qquad\qquad \Downarrow\ \text{control-}R_2^{-1}$$
$$|x_2\rangle \qquad\qquad \Downarrow\ H$$
$$\Downarrow$$
$$|x_1\rangle$$

2.3 The case $n = 3$: circuit C

$n = 3$
$w = (0.x_1x_2x_3)_2$

$$|\psi\rangle = \frac{1}{\sqrt{2^3}}\sum_{y=0}^{7} e^{2\pi i\,wy}\,|y\rangle \;=\; \begin{array}{c} \dfrac{|0\rangle + e^{2\pi i\,(0.x_3)_2}\,|1\rangle}{\sqrt{2}} \\ \otimes \\ \dfrac{|0\rangle + e^{2\pi i\,(0.x_2x_3)_2}\,|1\rangle}{\sqrt{2}} \\ \otimes \\ \dfrac{|0\rangle + e^{2\pi i\,(0.x_1x_2x_3)_2}\,|1\rangle}{\sqrt{2}} \end{array}$$

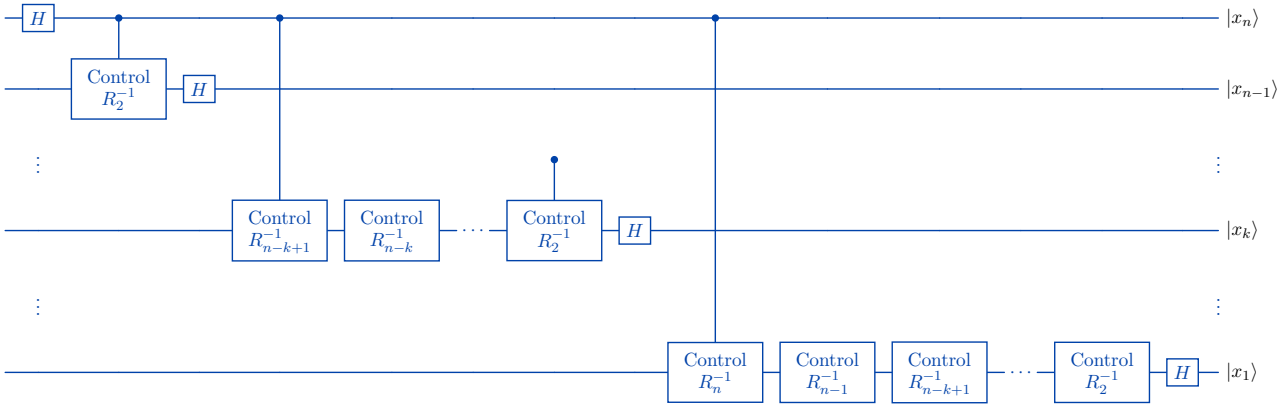


circuit C

2.4 General n: product form and circuit C

$w = (0.x_1x_2\cdots x_n)_2$

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i\, wy} |y\rangle \;=\; \begin{array}{c} \dfrac{|0\rangle + e^{2\pi i\,(0.x_n)_2}\,|1\rangle}{\sqrt{2}} \\ \otimes \\ \dfrac{|0\rangle + e^{2\pi i\,(0.x_{n-1}x_n)_2}\,|1\rangle}{\sqrt{2}} \\ \otimes \\ \vdots \\ \otimes \\ \dfrac{|0\rangle + e^{2\pi i\,(0.x_k x_{k+1}\cdots x_n)_2}\,|1\rangle}{\sqrt{2}} \\ \otimes \\ \vdots \\ \dfrac{|0\rangle + e^{2\pi i\,(0.x_1 x_2\cdots x_n)_2}\,|1\rangle}{\sqrt{2}} \end{array}$$



circuit C

2.5 Size and depth of the QFT_{2^n} circuit

size of QFT_{2^n} circuit:

$$\begin{aligned} &\#\text{ of } H \text{ gates} && = n \\ &\#\text{ of control gates} && = 1 + 2 + \cdots + (n-1) = \frac{n(n-1)}{2} = O(n^2) \\ &\text{Depth of circuit} && = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} = O(n^2) \end{aligned}$$

3 The Quantum Fourier Transform

3.1 Reversing circuit C

$w = (0.x_1x_2\cdots x_n)_2$ $\qquad$ $w = (0.x_1x_2\cdots x_n)_2$

$w = \dfrac{(x_1x_2\cdots x_n)_2}{2^n} = \dfrac{x}{2^n}$

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i\, wy} |y\rangle \quad \underset{\text{inverse of } C}{\overset{C}{\rightleftarrows}} \quad |x_1x_2\cdots x_n\rangle$$

we can do reverse here

there is a circuit

$$|x_1x_2\cdots x_n\rangle = \underbrace{|x_1x_2\cdots x_n\rangle}_{x} \xrightarrow{C^{-1}} |\psi\rangle\,, \qquad C^{-1}: \quad \mathrm{QFT}_{2^n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i \frac{x}{2^n} y} |y\rangle$$

Quantum Fourier Transform

3.2 QFT of order 2^n and of order m

QFT of order 2^n
$x = (x_1x_2\cdots x_n)_2$

$$\mathrm{QFT}_{2^n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i \frac{x}{2^n} y} |y\rangle \quad \longrightarrow$$

inverse QFT of order 2^n

$$\mathrm{QFT}_{2^n}^{-1}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{-2\pi i \frac{x}{2^n} y} |y\rangle$$

circuit of size, depth $O(n^2)$
exact calculation

Basis states are
$|0\rangle\,,\ |1\rangle\,,\ |2\rangle\,,\ \ldots,\ |2^n-1\rangle$

Basis states are
$|0\rangle\,,\ |1\rangle\,,\ |2\rangle\,,\ \ldots,\ |m-1\rangle\,, \quad m \neq 2^n$

QFT of order m

$$\mathrm{QFT}_m|x\rangle = \frac{1}{\sqrt{m}} \sum_{y} e^{2\pi i \frac{x}{m} y} |y\rangle$$

$$\mathrm{QFT}_m^{-1}|x\rangle = \frac{1}{\sqrt{m}} \sum_{y} e^{-2\pi i \frac{x}{m} y} |y\rangle$$

3.3 Approximate QFT when $m \neq 2^n$

if $m \neq 2^n$, $\mathrm{QFT}_m|x\rangle$ can only be calculated approximately by a circuit D

within error $\leq \dfrac{1}{T}$ $\longrightarrow$ $\max\limits_x \left\| \mathrm{QFT}|x\rangle - D|x\rangle \right\| \leq \dfrac{1}{T}$

with size of D polynomial in $\log m$ and $\log T$

4 Continuous and Discrete Fourier Transform

Continuous Fourier Transform $f(t) \longrightarrow F(\xi)$

$$F(\xi) = \int_{-\infty}^{\infty} f(t)\, e^{-2\pi i\, \xi t}\, dt$$

Discrete version

$$F_N = \sum_{k=0}^{N-1} f(k)\, e^{2\pi i}$$

5 No-Cloning Theorem (Wootters and Zurek, 1982)

$|\psi\rangle = \frac{1}{\sqrt{3}}\,|0\rangle + \sqrt{\frac{2}{3}}\,|1\rangle$

learn these approximately

$|\psi\rangle$ → n copies ← not possible

$$\underbrace{|\psi\rangle,\ |\psi\rangle,\ \ldots,\ |\psi\rangle}_{\text{measure them in standard basis}}$$

of 0 is n_0
of 1 is n_1

Law of Large numbers

$$\lim_{n\to\infty} \frac{n_0}{n} \longrightarrow \frac{1}{3} \qquad \lim_{n\to\infty} \frac{n_1}{n} \longrightarrow \frac{2}{3}$$

$|\psi\rangle$ is a 1-qubit state.
There is no unitary operator that can do this:

